CALL OF DUTY 4 LAN TOURNAMENT

2-Server Full Setup, Tournament Website & Broadcast Guide

CoD4x + Promod | S&D + TDM | Player Registration | Team Management | Admin Dashboard | RCON | Leaderboards | OBS Broadcast

Recommended production setup

PC 1 - Dedicated Game Server (2 server instances)
PC 2 - Tournament Admin / Control + Tournament Website
PC 3 - Broadcast / Spectator / OBS
Player PCs - Wired Gigabit LAN

Prepared as a practical event-day implementation and software architecture guide.

Version 2 - Expanded clear layout

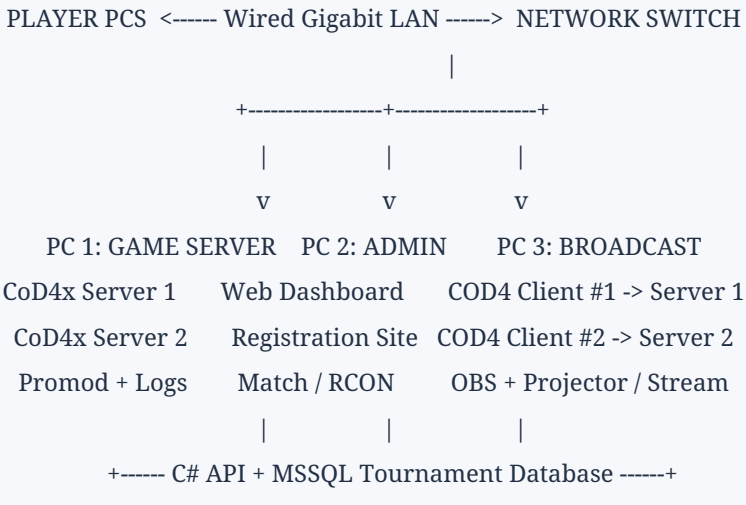
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1. Final Recommended Architecture

The recommended design separates game hosting, tournament administration, and broadcasting. The tournament website acts as the central source of truth for teams, players, matches, scores and leaderboards, while Promod handles the competitive in-game rules.

Overall Tournament Architecture



Core rule

Promod controls the game. The tournament website controls registrations, teams, matches, admin operations, results and leaderboards.

2. PC-wise Hardware Setup

2.1 PC 1 - Dedicated Game Server PC

This PC runs both CoD4x server instances. Keep it dedicated to server hosting during the event.

Item	Recommended
CPU	Modern 4-6 core CPU or better
RAM	16 GB or more
Storage	SSD
Network	Gigabit Ethernet

Item	Recommended
GPU	Not important for dedicated server use
Role	Server 1 + Server 2 only

Game Server Instances

Server 1 -> 192.168.10.10:28960

Server 2 -> 192.168.10.10:28961

Promod -> same tournament config on both

RCON -> separate strong password per server or managed securely

2.2 PC 2 - Tournament Admin / Control PC

This is the tournament control center. The admin should manage the event through a web dashboard instead of typing raw console commands for normal operations.

- Player and team registration
- Roster locking and validation
- Manual match creation and server assignment
- RCON server controls through GUI buttons
- Live server status and connected player list
- Result approval and manual override
- Team leaderboard and player leaderboard
- Broadcast overlay data and live match status

2.3 PC 3 - Broadcast / Spectator PC

One broadcast PC can monitor both servers by running two COD4 spectator/shoutcaster clients and capturing both in OBS.

Broadcast component	Purpose
COD4 Client #1	Spectate / shoutcast Server 1
COD4 Client #2	Spectate / shoutcast Server 2

Broadcast component	Purpose
OBS Studio	Capture, split-screen, scenes, overlays, recording / stream
Display Output	Projector / TV / LED screen

Best event setup

Use PC 2 only for administration. Use PC 3 only for spectator control and broadcast. This reduces mistakes during live matches.

3. Network & Server Addressing

Use a wired Gigabit switch for the server PC, admin PC, broadcast PC and all player PCs. Avoid Wi-Fi for competitive match traffic.

Device	Example static IP / port
Game Server PC	192.168.10.10
Admin PC	192.168.10.20
Broadcast PC	192.168.10.30
Server 1	192.168.10.10:28960
Server 2	192.168.10.10:28961

Player Connection Examples

/connect 192.168.10.10:28960

/connect 192.168.10.10:28961

4. Tournament Website - Main Modules

The tournament website should be the single place where organizers register participants, control matches and publish results. Keep the public pages separate from the admin-only pages.

Module	Main purpose
Public Registration	Player and team registration forms
Teams	View registered teams and approved rosters
Fixtures / Matches	Upcoming, live and completed matches
Team Leaderboard	Team standings for S&D / TDM
Player Leaderboard	Kills, deaths, assists, K/D and optional objective stats
Live Match	Current score, server, map, mode and player stats
Admin Dashboard	Tournament control center
RCON Control	Server actions through safe GUI buttons
Broadcast Overlay	Clean page for OBS browser source

Website Structure

PUBLIC WEBSITE

Registration | Teams | Fixtures | Leaderboards | Live Match

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ADMIN WEBSITE

Dashboard | Match Control | RCON | Results | Roster Lock

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C# API + MSSQL DATABASE

5. Player Registration Site

The registration page should collect enough information to identify a player and bind that player to one team. Nickname alone is not sufficient because it can be changed.

5.1 Registration Form Fields

Field	Required?	Notes
Full Name	Yes	Real participant name
Nickname / In-game Name	Yes	Displayed publicly
Mobile / Contact	Recommended	Used by organizer only
Team	Yes	Selected or entered during team registration
COD4 GUID / unique identifier	Recommended	Used for player validation where available
Role	Yes	Captain / Main / Substitute
Agreement	Yes	Tournament rules acceptance

5.2 Registration Rules

- A player can belong to only one team in the same tournament.
- A duplicate GUID or duplicate tournament player ID should be blocked.
- Registration can remain Pending until admin approval.
- After roster lock, changes require admin override and should be logged.
- Public pages should show nickname and team, not private contact details.

Player Registration Flow

Player submits registration

↓

Pending approval

↓

Admin verifies player / GUID / team

↓

Approved

↓

Player becomes part of locked team roster

6. Team Registration & Roster Management

A team page allows a captain or admin to build the roster. For a 5v5 format, five main players plus one optional substitute is practical.

Team field	Example
Team Name	ALPHA
Team Code	ALP
Captain	KAZ
Main Players	5
Substitutes	0-1 or as tournament rules allow
Roster Status	Draft / Approved / Locked
Team Status	Active / Withdrawn / Disqualified

6.1 Roster Validation

Example Team Roster

TEAM ALPHA

KAZ - Captain ✓ Approved

NIMAL - Main ✓ Approved

DEV - Main ✓ Approved

MAX - Main ✓ Approved

ZERO - Main ✓ Approved

SAM - Substitute ✓ Approved

Important

When Match ALPHA vs BRAVO is active, only approved ALPHA and BRAVO roster members should be accepted as valid players for that match. Any player registered to another team must be flagged.

7. Admin Dashboard

The admin dashboard is the most important web screen during the tournament. It should show the event state at a glance and keep the main actions one or two clicks away.

7.1 Dashboard Summary Cards

Card	Example information
Teams	16 Registered / 16 Approved
Players	90 Registered / 86 Approved
Matches	12 Completed / 2 Live / 10 Upcoming
Server 1	Online - ALPHA vs BRAVO
Server 2	Online - DELTA vs ECHO
Pending Approvals	4 Players
Warnings	1 roster mismatch

7.2 Main Admin Navigation

Admin Menu

DASHBOARD
├─ Teams
├─ Players
├─ Matches
├─ Servers / RCON

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├─ Live Matches
├─ Results
├─ Team Leaderboard
├─ Player Leaderboard
└─ Settings / Audit Log
```

7.3 Suggested Dashboard Layout

Area	What the admin sees
Top Bar	Tournament name, admin profile, current time, emergency stop / logout
Server Panel	Server 1 and 2 online state, current match, player count
Live Match Panel	Team names, score, mode, map, status
Pending Tasks	Player approvals, roster issues, missing results
Quick Actions	Create Match, Assign Server, Restart Map, Send Message
Leaderboard Preview	Top teams and top players

8. Match Management

Matches are created manually by the admin. This matches your tournament plan and avoids unnecessary fixture automation.

8.1 Create Match Form

Field	Example
Team A	ALPHA
Team B	BRAVO

Field	Example
Mode	S&D or TDM
Map	mp_crash
Server	Server 1
Start Time	Manual / scheduled
Match Format	Single map / Best of 3
Status	Draft / Ready / Live / Completed / Cancelled

Match Lifecycle

Admin creates match
↓
Assign Server 1 or Server 2
↓
Select S&D / TDM + map + config
↓
Teams join
↓
Roster validation
↓
Start via RCON GUI
↓
Live match
↓
Result captured / verified
↓
Leaderboard updates

9. RCON / Server Control Panel

RCON remains the technical control mechanism, but the admin should normally use dashboard buttons. The backend sends the correct RCON command to the selected server.

GUI Action	Admin purpose
Refresh Players	Show connected player list
Change Map	Load selected map
Load S&D Config	Prepare Search & Destroy rules
Load TDM Config	Prepare Team Deathmatch rules
Restart Match	Restart current map / match
Kick Player	Remove selected player
Send Message	Broadcast admin message to server
Server Password	Set / clear join password
Stop / Reset	Emergency control

Security

Never expose the RCON password in the public website, browser JavaScript, OBS overlay or screenshots. Store it server-side only.

10. Search & Destroy + Team Deathmatch Rules

10.1 Search & Destroy (S&D)

- Promod competitive settings.
- Track rounds won and lost.
- Use round difference as a tie-breaker where required.
- Broadcast can show round score, timer, alive/dead state and bomb status.

10.2 Team Deathmatch (TDM)

- Use one fixed score limit and/or time limit across both servers.
- Track Score For, Score Against and Score Difference.
- Player leaderboard can emphasize kills, deaths, assists and K/D.
- Use the same TDM config on both servers.

10.3 Same Tournament, Two Modes

Store the mode on every match. The result fields and leaderboard logic can then adapt based on S&D or TDM.

Mode	Primary result	Useful tie / stat data
S&D	Rounds won	Round difference, rounds for, head-to-head
TDM	Final team score	Score difference, total score for, head-to-head

11. Leaderboard Management

11.1 Team Leaderboard

The team leaderboard should update after a completed result is approved. Store raw match results and calculate standings from those results so corrections are easy.

Pos	Team	P	W	L	For	Against	Diff	Pts
1	ALPHA	5	5	0	65	39	+26	15
2	BRAVO	5	4	1	59	42	+17	12
3	DELTA	5	3	2	54	47	+7	9

Recommended base points: Win = 3, Loss = 0. If you use draws in any mode, define the draw value before the event.

Recommended ranking order

1. Points
2. Head-to-head result
3. Score / Round Difference
4. Score / Rounds For
5. Decider match if still required

11.2 Player Leaderboard

Pos	Player	Team	Kills	Deaths	Assists	K/D
1	KAZ	ALPHA	89	41	18	2.17
2	DEV	BRAVO	76	48	21	1.58
3	ZERO	ALPHA	61	42	17	1.45

Optional extra player stats: plants, defuses, headshots, accuracy, damage and matches played - only include stats that can be collected reliably.

11.3 Result Approval and Manual Override

Safe Result Workflow

Automatic / imported result

↓

Admin reviews score

↓

APPROVE RESULT

↓

Leaderboard recalculated

↓

If correction needed -> Admin Override -> Audit Log

12. Live Match & Broadcast Website

Create a dedicated live-match page and a clean broadcast-overlay page. The live page is for viewers; the overlay page is designed to be added into OBS as a browser source.

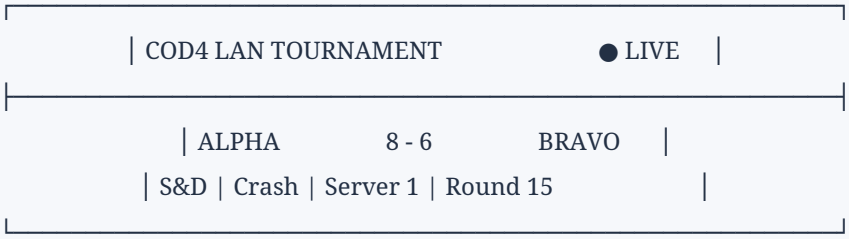
12.1 Live Match Page

Area	Display
Header	Tournament name + LIVE indicator
Match	ALPHA vs BRAVO

Area	Display
Score	Current S&D rounds or TDM score
Info	Server, mode, map, match format
Players	Current team players and key stats
Status	Ready / Live / Paused / Completed

12.2 OBS Overlay Page

Example Broadcast Overlay



The overlay should have a transparent or simple background, large text, and no admin controls. OBS can load it as a Browser Source.

13. Broadcast PC / OBS Setup

The broadcast PC runs two COD4 spectator clients. OBS captures both and allows the operator to switch between full-screen Server 1, full-screen Server 2, and a 2-way split view.

OBS Scene	Contents
Scene 1 - Server 1	COD4 Client #1 + live overlay
Scene 2 - Server 2	COD4 Client #2 + live overlay
Scene 3 - Split Screen	Server 1 and Server 2 side-by-side
Scene 4 - Leaderboard	Browser source showing team leaderboard

OBS Scene	Contents
Scene 5 - Break / Next Match	Tournament logo + upcoming matches

Broadcast Flow

COD4 Client #1 -> Server 1 spectator
COD4 Client #2 -> Server 2 spectator
↓
OBS Studio
/ | \
Server 1 Server 2 Split View
↓
Projector / Stream

Performance note

Two simultaneous COD4 spectator clients plus OBS is practical on a reasonably strong PC. Keep graphics settings moderate if the broadcast PC struggles.

14. Database & API Structure

A simple ASP.NET Core API + Microsoft SQL Server backend fits this system well. The database should keep raw entities and raw match results; leaderboards can be calculated from them.

14.1 Suggested Tables

Table	Purpose
Tournaments	Event settings and status
Teams	Team identity and status
Players	Player identity and public nickname
TeamPlayers	Team membership and role
Servers	Server IP, port, status and secure RCON reference

Table	Purpose
Matches	Teams, mode, map, server, status
MatchResults	Final verified scores and winner
PlayerMatchStats	Per-match player stats
AdminUsers	Admin accounts and roles
AuditLogs	Roster, result and control changes

14.2 Suggested API Groups

API group	Examples
/api/players	register, approve, list, lock
/api/teams	create, roster, approve, lock
/api/matches	create, assign server, start, complete
/api/servers	status, players, rcon actions
/api/leaderboards	team standings, player rankings
/api/live	current matches and overlay data
/api/admin	settings, audit, manual overrides

15. Full Tournament-Day Workflow

6. Power on network switch, Server PC, Admin PC and Broadcast PC.
7. Start CoD4x Server 1 and Server 2 and confirm both are online.
8. Open Admin Dashboard and verify server status / RCON access.
9. Open two COD4 spectator clients on Broadcast PC and connect one to each server.

10. Open OBS and test Server 1, Server 2, Split Screen and Leaderboard scenes.
11. Approve any pending player or team registrations before matches begin.
12. Create the next match manually and assign Server 1 or Server 2.
13. Confirm mode, map and server config.
14. Players join; admin validates the roster.
15. Start match through RCON GUI.
16. Broadcast operator selects the best spectator view.
17. At match end, result is captured or entered, then admin approves it.
18. Team and player leaderboards update.
19. Prepare the next match while the completed match is archived.

16. Security, Backup & Failure Plan

- Keep RCON passwords server-side and change them after the event if shared with staff.
- Use admin login roles; do not allow public users to access result edit or server control pages.
- Back up the tournament database before the event and after major match blocks.
- Keep manual result entry available even if automatic log/stat collection fails.
- Keep printed or offline copies of team rosters and the match schedule as a fallback.
- If one server fails, move the match to the other server and update the match server assignment in the dashboard.
- If broadcast fails, the game servers and admin system must continue independently.

17. Build Priority Checklist

For a fast implementation, build in this order so the tournament can still run even if advanced automation is unfinished.

Priority	Feature	Must be ready?
1	Teams + Players + Registration + Approval	YES
2	Roster lock / duplicate player prevention	YES
3	Manual Match Creation + Server Assignment	YES
4	Result Entry + Team Leaderboard	YES
5	Admin Dashboard + Server status	YES
6	RCON GUI buttons	Highly recommended

Priority	Feature	Must be ready?
7	Live Match page + OBS overlay	Highly recommended
8	Player leaderboard	Recommended
9	Automatic player validation / log integration	Add if stable
10	Advanced automatic stats	Optional after core system works

Final recommended setup

PC 1: 2 CoD4x + Promod servers. PC 2: registration, admin dashboard, match control, RCON and leaderboards. PC 3: two spectator clients + OBS + projector/stream. Use a wired Gigabit LAN and keep manual result override as the tournament-day safety net.